

Mageean Brown

📍 Pasadena, CA ✉️ mrbrown@caltech.edu 🌐 mageeanbrown.github.io in mageean-brown-32279921b

EDUCATION

- California Institute of Technology, M.S. in Aeronautics** *Sept 2024 – June 2025*
Graduate Aerospace Laboratories (GALCIT). Pasadena, CA. GPA: 3.9/4.0
- Track in Fluid Dynamics and Control Systems
- Brown University, B.S. with Honors in Mechanical Engineering** *Jan 2021 – May 2024*
School of Engineering. Providence, RI. GPA: 3.9/4.0
- Selected Courses: Renewable Energy Technologies, Advanced Engineering Mechanics, Material Science

TECHNICAL SKILLS

- **Programming & Tools:** MATLAB, Python, Arduino, Git, Zoho CRM, Microsoft Office, Google Suite, Slack
- **Design:** SolidWorks (CAD, FEA, CFD, tolerance stack analysis, thermal analysis), Fusion 360 (CAD, CAM, Modal Analysis), Onshape, AutoCAD, drawings using GD&T and ASME Y14.5 standards
- **Manufacturing:** 3D printing (FDM and SLA), machining (manual and CNC), TIG welding, soldering, sheet metal fabrication, injection molding
- **Laboratory & Analysis:** Particle image velocimetry (PIV), CFD (MATLAB and SolidWorks), particle shadow velocimetry, PIV processing in PIVlab and DaVis, dye visualization, force analysis (with transducer and optical methods), ImageJ

RELEVANT EXPERIENCE

- NuWatt Energy** *Woburn, MA*
Project Manager Intern *Jan 2022–April 2022*
- Moved solar projects from start to finish by filling out permits, communicating with subcontractors to organize project logistics, and coordinating with clients through Zoho CRM.
 - Learned relevant energy codes, ASHRAE standards, and incentives for installing solar.
 - Helped manage the timelines of up to five concurrent projects to reach all project deadlines.

PROJECTS

Full project portfolio available on my website: <https://mageeanbrown.github.io/>

Brown Formula Racing

- Led design and fabrication of the chassis (2024) and steering system (2023) for student-built race cars. Both cars completed all events at the FSAE Michigan international competition.
- Validated chassis designs with finite element analysis (FEA) in SolidWorks, applied physical tests on the chassis to determine torsional stiffness, and used piezoelectric vibration sensors to ensure safe coupling of the engine and chassis.
- Taught welding and design skills by guiding 5 new members through an intro manufacturing project and held weekly welding hours to help members practice tungsten inert gas (TIG) welding.
- Took a 6-week machining course and subsequently accumulated hundreds of hours of machining experience using lathes and mills (manual and CNC).

Hand Tremor Reduction Device

- Executed the full product development cycle of an electro-mechanical wearable device to reduce the amplitude of hand tremors by about 40%, targeted to alleviate Parkinson's Disease symptoms.
- Implemented a closed-loop feedback PID control system in Python and Arduino to modulate the motion of solenoids in the device to decrease tremor amplitude, which was measured with an IMU.

High Speed Pipe Flow Imaging

- Studied the laminar-to-turbulent flow transition in pipe-flow applications via optical techniques, namely particle

image velocimetry (PIV) for flow field quantification.

- Fabricated a high-precision translation stage (0.005"/rev, 6" total travel) to position laser sheet forming lenses at precise distances.

Drone Flight Control System

- Designed and simulated a linear quadratic Gaussian (LQG) control system to maintain the angular rates, position, and velocity of a drone model using MATLAB and Simulink.
- Achieved a minimum snap trajectory between 100 waypoints, which ensured smooth and safe trajectories in obstacle-dense spaces.
- Implemented and tuned a Kalman filter to estimate 12 states from IMU, GPS, and compass data.

Bicycle Framebuilding

- Independently designed, validated, and welded a custom steel bicycle frame from scratch.
- Used manual and CNC machines to fabricate a 24-part welding jig from aluminum within tolerances of 0.002".

RESEARCH

Dabiri Lab

Graduate Researcher

Caltech

June 2025 – Present

- Studying vortex interactions between tandem bio-hybrid (robotically-controlled) jellyfish to be used for ocean exploration and monitoring.
- Design and program circuitry for jellyfish swimming control using Arduino, which can stimulate the contraction of jellyfish muscles and increase swimming speeds by approximately 4x natural swimming speeds.

Wilhelmus Lab

Undergraduate Researcher

Brown University

Feb 2023 – Aug 2024

- Applied knowledge of fluid mechanics to study shrimp transport for the development of bio-inspired autonomous underwater vehicles (AUVs).
- Designed and built modular robotic shrimp swimming appendages (pleopods) that accurately simulated metachronal swimming based on biological studies of shrimp kinematics using MATLAB and Arduino.
- Studied the effects of pleopod geometry on swimming performance using 3 different experimental techniques: particle image velocimetry (PIV), force measurement, and kinematic tracking in MATLAB.

Witman Lab

Undergraduate Researcher

Brown University

Sept 2021 – Jan 2024

- Processed over 1000 hours of underwater acoustic data using Triton software in MATLAB to determine the impact of anthropogenic activity on marine ecosystems in the Galápagos Islands.
- Helped organize outreach events to share marine ecology research with younger students and improve the accessibility of marine science to the general public. Projects include a month-long Marine Ecology Research Exhibition at the Granoff Center for the Creative Arts.

PUBLICATIONS

- Oliveira Santos, S., **Brown, M.**, Kim, M., Tack, N., Wilhelmus, MM. Force generation in a bio-inspired rotating and pitching plate of variable area. *Under prep for Journal of Fluid Mechanics; manuscript available upon request.*

Conference Proceedings:

- Oliveira Santos, S., **Brown, M.**, Kim, M., Tack, N., Wilhelmus, MM. Analysis of Shrimp Appendage Cupping on Swimming Performance through a Bio-inspired Model. (Oral Presentation) APS 77th Annual Meeting of the Division of Fluid Dynamics, Salt Lake City, UT.
- **Brown, M.**, Oliveira Santos, S., Tack, N., Wilhelmus, MM. Analysis of Shrimp Appendage Cupping on Swimming Performance through a Bio-inspired Model. (Oral Presentation) APS 76th Annual Meeting of the Division of Fluid Dynamics, Washington, DC.
- **Brown, M.**, Oliveira Santos, S., Tack, N., Wilhelmus, MM. Appendage cupping and hydrodynamics of shrimp-inspired model. (Poster Presentation) 2023 Summer Research Symposium, August 4th, Providence, RI.

TEACHING

The Wheeler School

Computer Aided Design Teacher

*Providence, RI
March 2023 – May 2024*

- Developed an introductory course in computer aided design (CAD) entirely from scratch using Onshape.
- Prepared and taught weekly classes to middle-school students, organized projects, and incorporated hands-on activities, such as lessons using 3D-printers.

Brown University

Undergraduate Teaching Assistant

*Woburn, MA
Jan 2022–April 2022*

- Teaching assistant for 3 courses: Fluid Mechanics (ENGN0810), Fluid Mechanics Through Hovercraft Physics (CEEN0926), Introduction to Engineering (ENGN0030).
- Held weekly office hours to help students with course material and homework, led laboratory sessions, and graded homework.

HONORS AND AWARDS

- **NSF GRFP Honorable Mention** (National Science Foundation) *Spring 2025*
- **Domenico A. Ionata '26 Award** (Brown University) *Spring 2024*
- **Undergraduate Teaching and Research Award** (Brown University) *Fall 2023*
- **Research at Brown (RAB) Grant** (Brown University) *Fall 2023*
- **APS Travel Grant** (American Physical Society) *Fall 2023*
- **Valedictorian** (Hoosick Falls Central School) *June 2020*